

LECTURE 4

Topics

Trade-off between risk and risk premium, Investment opportunity set, Minimum-variance frontier, Efficient frontier, Capital allocation line, Optimal portfolio allocation, Capital market line

- In Lecture 3 we noted that investors use an asset's historical distribution of monthly returns to learn about:
 - The asset's *expected* return over the next month
 - The asset's return volatility (or the asset's total risk) over the next month
- T-Bills have the unique feature that their interest rates are pre-determined for the next 30 days and there is virtually no risk in those rates changing (it will take a US government default for that to happen)
- Thus, T-Bill's return volatility over the next month is virtually zero
- We think of T-Bills as being effectively risk-free, and we call T-Bills a *risk-free asset*
- Other money market instruments that we discussed above also qualify as candidates for risk-free assets

- Investors are generally risk averse, and they prefer less risk to more risk, everything else equal
- Except for the T-Bills, most assets will expose investors to risk
- It is then natural for the investor to compare the expected return of a risky asset, $E(r)$, to the expected returns of the T-Bills, r_F
- The risk premium of an asset captures exactly this difference:

$$\text{Risk premium} = E(r) - r_F$$

- Risk premium is generally viewed as compensation for bearing risk
- In particular, a risk averse investor would consider investing in a risky asset only if it has a positive risk premium
- The actual amount invested in the risky asset will depend not only on the risk premium but also the risk of the asset and the degree of investor's risk aversion

- In economics, consumers form preferences over goods and services that they would like to consume now or in the future
- Because financial assets facilitate future consumption, consumers also form preferences over the return distributions of these assets
- In particular, consumers care about the mean return and the risk of financial assets
- In economics, a consumer's preference over the mean return and the risk of financial assets can be quantified using *utility* functions
- These utility functions take as inputs various statistics of the return distribution of an asset (e.g., mean return, return volatility) and output a "happiness" score for the asset
- Utility functions allow consumers to rank investment opportunities based on their happiness score

- We illustrate the concept of utility function with an example

Assumption on return distributions

Suppose that the returns of all financial assets are normally distributed

- Because normal distributions are fully characterized by their mean and standard deviation, the return distributions will be fully characterized by their mean return, $E(r)$, and return volatility, σ

Utility function

The utility function, U , calculates the "happiness" score for an asset with mean return, $E(r)$, and return volatility, σ as follows:

$$U = E(r) - \frac{1}{2}A\sigma^2$$

The parameter A is the consumer's *degree of risk aversion*

- Note the dependency of the happiness score, U , on the risk aversion parameter A
- $A > 0$ corresponds to a risk averse investor, who only considers risky investment opportunities ($\sigma > 0$) if they have large risk premiums, and thus large expected returns $E(r)$
- $A = 0$ corresponds to a risk neutral investor, who doesn't care about the risk, but cares about the expected return of an investment opportunity
- $A < 0$ corresponds to a risk lover investor, who prefers riskier investment opportunities

- We now use the the utility function and happiness scores to rank among investment opportunities
- Suppose the risk-free asset has a mean return of $r_F = 2\%$
- Consider the following investment opportunities L , M and H
 - L: expected return $E(r_L) = 4\%$, return volatility $\sigma_L = 5\%$
 - M: expected return $E(r_M) = 6\%$, return volatility $\sigma_M = 10\%$
 - H expected return $E(r_H) = 11\%$, return volatility $\sigma_H = 20\%$
- The risk premiums are respectively :
 - $E(r_L) - r_F = 4\% - 2\% = 2\%$
 - $E(r_M) - r_F = 6\% - 2\% = 4\%$
 - $E(r_H) - r_F = 11\% - 2\% = 9\%$

- We rank these opportunities based on their happiness scores and choose the one with the largest score
- For an investor with risk aversion $A = 2$, opportunity H is the most attractive:
 - $U_L = E(r_L) - \frac{1}{2}A\sigma_L^2 = 4\% - \frac{1}{2} \times 2 \times (5\%)^2 = 0.0375$
 - $U_M = E(r_M) - \frac{1}{2}A\sigma_M^2 = 6\% - \frac{1}{2} \times 2 \times (10\%)^2 = 0.05$
 - $U_H = E(r_H) - \frac{1}{2}A\sigma_H^2 = 11\% - \frac{1}{2} \times 2 \times (20\%)^2 = 0.07$
- For an investor with risk aversion $A = 5$, opportunity M is the most attractive:
 - $U_L = 4\% - \frac{1}{2} \times 5 \times (5\%)^2 = 0.0338$
 - $U_M = 6\% - \frac{1}{2} \times 5 \times (10\%)^2 = 0.035$
 - $U_H = 11\% - \frac{1}{2} \times 5 \times (20\%)^2 = 0.01$

- Utility functions allow investors to rank among investment opportunities
- The optimal choice depends on the risk and mean return of the opportunity but also the risk aversion of the investor
- A more risk averse investor will be happier with a less risky investment opportunity
- Furthermore, we have the following result

Mean-variance criterion

If two investment opportunities, A and B, are such that:

$$E(r_A) \geq E(r_B), \text{ and}$$

$$\sigma_A \leq \sigma_B,$$

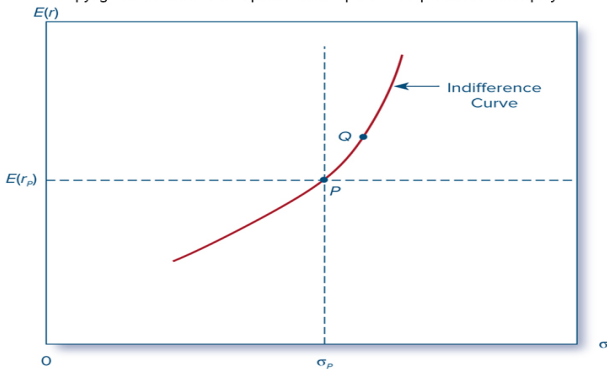
then a risk averse investor will always choose A over B.

- When two investment opportunities, P and Q , generate the same happiness score for an investor,

$$U_P = U_Q,$$

then P and Q lie on the same *utility indifference curve*

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- We will identify a portfolio by the weights of its assets
- We illustrate how to identify these weights with an example
- Suppose that an investor has a portfolio, P , that consists of:
 - \$90,000 in the risk-free asset, F
 - \$210,000 in a risky portfolio, of which:
 - \$113,400 in stocks, S
 - \$96,600 in bonds, B
- The total net worth of the portfolio is \$300,000
- Then the weights of the assets in portfolio P are:
 - Risk free asset: $w_F = 90,000/300,000 = 0.3$
 - Stocks: $w_S = 113,000/300,000 = 0.378$
 - Bonds: $w_B = 96,600/300,000 = 0.322$
- We identify portfolio P with the set of weights (w_F, w_S, w_B)
- Note that these weights add up to 1

- Sometimes it's convenient to group the stocks, S , and the bonds, B , into a risky portfolio P^*
- This way, we can view portfolio P as being invested \$90,000 in the risk-free asset, F , and \$210,000 in the risky portfolio, P^*
- The weights of portfolio P become:
 - Risk-free asset: $w_F = 90,000/300,000 = 0.3$
 - Risky portfolio: $w_{P^*} = 210,000/300,000 = 0.7$
- The weights of portfolio P^* are:
 - Stocks: $w_S^* = 113,000/210,000 = 0.54$
 - Bonds: $w_B^* = 96,600/210,000 = 0.46$

- We now analyze an investor's optimal portfolio allocation in the simpler case of just two assets: one risk-free and the other risky
- We assume that:
 - Risk-free asset, F , has return r_F and no volatility, $\sigma_F = 0$
 - Risky asset, P , has expected return $E(r_P)$ and volatility σ_P
 - Covariance between P and F is $\sigma_{PF} = 0$ (F has no risk)
- The covariance, σ_{PF} , between P and F captures the extent to which the two assets move in sync (i.e., how often they go up at the same time, or go down at the same time)
- Because the risk-free asset F never moves up or down (because it has no risk), while asset P does, the two assets never move in sync, and $\sigma_{PF} = 0$

Optimal portfolio allocation

Optimal portfolio of assets F and P is the portfolio that generates the highest happiness score

- The first step in the analysis is to find the set of all possible portfolios C that we can build from the assets P and F
- This set of portfolios is called the *investment opportunity set*
- To simplify the analysis, we assume some values for the return statistics of the two assets:
 - $r_F = 2\%$ and $\sigma_F = 0$
 - $E(r_P) = 10\%$ and $\sigma_P = 22\%$ and $\sigma_{P,F} = 0$
- Consider now a portfolio C with weights y and $1 - y$
- We want to calculate the expected return and the return volatility of portfolio C

Expected return of a portfolio of two assets

If portfolio C has a weight of y on asset P and a weight of $1 - y$ on asset F , then the expected return of portfolio C can be calculated as:

$$E(r_C) = y \times E(r_P) + (1 - y) \times E(r_F)$$

- Thus, the expected return of portfolio C becomes:

$$E(r_C) = y \times E(r_P) + (1 - y) \times r_F$$

$$E(r_C) = y \times 10\% + (1 - y) \times 2\%$$

$$E(r_C) = 2\% + 8\%y$$

Variance and volatility of a portfolio of two assets

If portfolio C has a weight of y on asset P and a weight of $1 - y$ on asset F , then the variance of portfolio C can be calculated as:

$$\sigma_C^2 = y^2 \times \sigma_P^2 + (1 - y)^2 \times \sigma_F^2 + 2y(1 - y) \times \sigma_{P,F}$$

while the volatility of portfolio C is simply the square root of the variance:

$$\sigma_C = \sqrt{\sigma_C^2}$$

- σ_P^2 is the variance of asset P ,
- σ_F^2 is the variance of asset F , and
- $\sigma_{P,F}$ is the covariance of asset P and F ,

- Recall that the risk-free asset, F , has no risk, and so $\sigma_F = 0$ and $\sigma_{P,F} = 0$
- The variance of portfolio C becomes:

$$\sigma_C^2 = y^2 \times \sigma_P^2 + (1 - y)^2 \times 0 + 2y(1 - y) \times 0$$

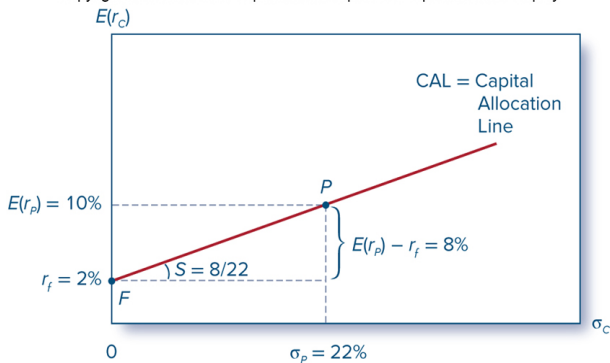
$$\sigma_C^2 = y^2 \times \sigma_P^2 = y^2 \times (22\%)^2$$

- While portfolio volatility is:

$$\sigma_C = \sqrt{\sigma_C^2} = 22\%y$$

- Note that both the expected return and the return volatility of portfolio C depend linearly on y
- Thus, the investment opportunity set is a line that passes through the assets F and P
- This line is sometimes called the *capital allocation line* (CAL)

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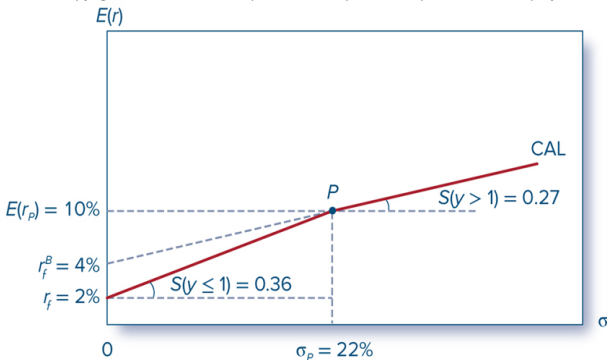
- The CAL's intercept is the risk-free rate, $r_f = 2\%$
- The CAL's slope is the Sharpe ratio of asset P:

$$\text{slope} = \frac{10\% - 2\%}{22\%} = 0.36$$

- The portfolios on CAL between F and P invest a positive amount in both F and P :
 - $1 \geq y \geq 0$
 - $1 - y \geq 0$
- The portfolios on CAL above P invest more than 100% in P and thus a negative amount in F
 - $y > 1$
 - $1 - y < 0$
- These later portfolios are called *leveraged* portfolios, because we borrow at the risk-free rate to over invest in the risky asset
- If we cannot borrow at the risk-free rate r_F , but we can borrow at a higher rate, say $r_B > r_F$, then the investment opportunity set changes

- Investment opportunity set with borrowing rate r_B

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- Portfolios that invest positive amounts in F and B lie on the CAL from above
- However, leveraged portfolios lie on the line connecting the borrowing rate r_B to P

- Now that we know the investment opportunity set, we need to pick the one portfolio in this set that achieves the highest happiness score for our investor
- There are two ways in which we can do this
 - Geometric
 - Calculus
- Suppose our investor has a risk aversion of $A = 4$
- Then, for a portfolio C , the happiness score will be calculated using the utility formula that we discussed above

$$U(C) = E(r_C) - \frac{1}{2} \times A \times \sigma_C^2$$

- The first approach is to calculate happiness scores for different portfolios C and then plot them

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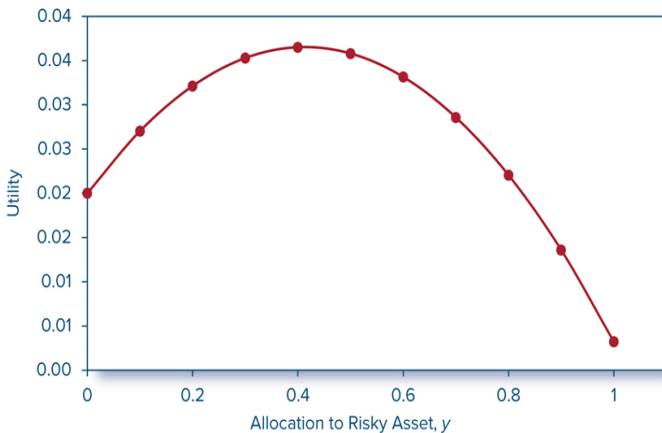
Table 6.4

Utility levels for various positions in risky assets (y) for an investor with risk aversion $A = 4$. The risky portfolio has a risk premium of 8% and a standard deviation of 22%. The risk-free rate is 2%.

(1) y	(2) $E(r_C)$	(3) σ_C	(4) $U = E(r) - \frac{1}{2}A\sigma^2$
0	0.020	0	0.0200
0.1	0.028	0.022	0.0270
0.2	0.036	0.044	0.0321
0.3	0.044	0.066	0.0353
0.4	0.052	0.088	0.0365
0.5	0.060	0.110	0.0358
0.6	0.068	0.132	0.0332
0.7	0.076	0.154	0.0286
0.8	0.084	0.176	0.0220
0.9	0.092	0.198	0.0136
1.0	0.100	0.220	0.0032

- We now plot the happiness scores U against the weight y

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- Portfolio $C^* = (0.4, 0.6)$ achieves the highest happiness for our investor with risk aversion $A = 4$

- An alternative to identify the optimal portfolio is to use calculus
- First recall that for a portfolio C , the expected return and return volatility are:
 - $E(r_C) = 2\% + 8\%y$
 - $\sigma_C = 22\%y$
- We now plug these expressions into the utility formula:

$$U(C) = E(r_C) - \frac{1}{2} \times A \times \sigma_C^2$$

$$U(y) = 2\% + 8\%y - \frac{1}{2} \times 4 \times (22\%y)^2$$

$$U(y) = 2\% + 8\%y - \frac{1}{2} \times 4 \times (22\%)^2 y^2$$

- We want the value y^* that maximizes U

- A quick way to find y^* is through the use of Calculus
- Recall from FNCE 2010 that y^* is the solution to the equation:

$$U'(y) = 0$$

where $U'(y)$ is the derivative of function $U(y)$.

- Next we focus on calculating $U'(y)$
- Recall that:
 - The derivative of a constant is always 0
 - $y' = 1$
 - $(y^2)' = 2y$

- We are now ready to calculate $U'(y)$:

$$U'(y) = (2\%)' + 8\% \times y' - \frac{1}{2} \times 4 \times (22\%)^2 \times (y^2)'$$

$$U'(y) = 0 + 8\% \times 1 - \frac{1}{2} \times 4 \times (22\%)^2 \times 2y$$

$$U'(y) = 8\% - \frac{1}{2} \times 4 \times (22\%)^2 \times y$$

- We now solve for y^* in equation: $U'(y) = 0$

$$8\% - \frac{1}{2} \times 4 \times (22\%)^2 y = 0$$

$$y^* = \frac{8\%}{4 \times (22\%)^2} = 41.3\%$$

- The optimal allocation in the risky asset, y^* , depends on:
 - Risk premium of the risky asset:

$$E(r_P) - r_F = 10\% - 2\% = 8\%$$

- The return volatility of the risky asset: $\sigma_P = 22\%$
 - Risk aversion of the investor: $A = 4$
- We can make a more general formula out of this calculation:

Optimal portfolio allocation with one risky asset

An investor with risk aversion A that wants to invest in the risky asset, P , and risk-free asset, F , will allocate y in P and $1 - y^*$ in F , where y^* is given by:

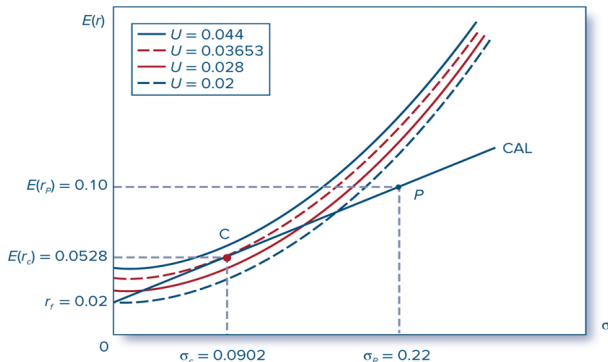
$$y^* = \frac{E(r_P) - r_F}{A \times \sigma_P^2}$$

- Plugging $y^* = 41.3\%$ back in the formula for U , we get:

$$U(y) = 2\% + 8\% \times y - \frac{1}{2} \times 4 \times (22\%)^2 \times y^2 = 0.03653$$

- Geometrically, this is the highest utility level of any of the portfolios in the investment opportunity set

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- We now explore the case when the investor wants to invest in the risk-free asset, F , and two risky assets, P and Q
- Having a second risky asset introduces a extra layer of complexity because the two risky assets co-move together
- The co-movement between P and Q is measured by *covariance*:

$$\sigma_{PQ} = \frac{(r_1^P - E(r_P))(r_1^Q - E(r_Q)) + \dots + (r_N^P - E(r_P))(r_N^Q - E(r_Q))}{N}$$

where

- $E(r_P)$ is the historic mean return of asset P
- $E(r_Q)$ is the historic mean return of asset Q
- N is the length of the historical time series for P and Q
- Like the return volatility, the covariance captures the return deviations of the two assets from their mean returns

- We start by describing the investment opportunity set generated by the two risky assets P and Q
- Suppose that the two risky assets have the following characteristics:
 - P has expected return $E(r_P)$ and return volatility of σ_P
 - Q has expected return $E(r_Q)$ and return volatility of σ_Q
 - Return covariance between P and Q is σ_{PQ}
- The investment opportunity set generated by P and Q is the set of all portfolios that invest in P and Q only
- Recall that we can identify a portfolio S by its weights: (w_P, w_Q)
- Clearly, weights must always add up to 1: $w_P + w_Q = 1$
- We now want to calculate the expected return and the return volatility of the portfolio (w_P, w_Q)

- We will introduce some new notations and operations that would simplify considerably our analysis
- Matrices are characterized by *rows X columns*
 - Arrange the expected returns $E(r_P)$ and $E(r_Q)$ in a matrix with one row and two columns (a 1×2 matrix):

$$E(r) = (E(r_P), E(r_Q))$$

- Arrange the return volatilities and return covariances in a matrix with two rows and two columns (a 2×2 matrix):
- $$\Sigma = \begin{pmatrix} \sigma_P^2 & \sigma_{PQ} \\ \sigma_{QP} & \sigma_Q^2 \end{pmatrix}$$
- Finally, arrange the portfolio weights w_P and w_Q in a matrix with one row and two columns (a 1×2 matrix):

$$w = (w_P, w_Q)$$

- Common Matrix operations include *transpose* and *multiplication*
- Transpose: Transforms a 1×2 matrix into a 2×1 matrix
 - For example, the transpose of the 1×2 matrix $w = (w_P, w_Q)$ is a 2×1 matrix denoted with w^T :

$$w^T = \begin{pmatrix} w_P \\ w_Q \end{pmatrix}$$

- Multiplication: We can multiply matrix A with matrix B if:

of columns of matrix A = # of rows of matrix B

- For example, we can multiply matrix Σ by matrix w^T :

$$\Sigma \times w^T = \begin{pmatrix} \sigma_P^2 & \sigma_{PQ} \\ \sigma_{QP} & \sigma_Q^2 \end{pmatrix} \times \begin{pmatrix} w_P \\ w_Q \end{pmatrix} = \begin{pmatrix} \sigma_P^2 \times w_P + \sigma_{PQ} \times w_Q \\ \sigma_{QP} \times w_P + \sigma_Q^2 \times w_Q \end{pmatrix}$$

- Using matrices and matrix operations, we can now calculate quite fast the expected return and the return volatility of the portfolio S with weights $w_S = (w_P, w_Q)$
- Expected return of portfolio S is:

$$E(r_S) = E(r) \times w_S^T$$

- Return variance of portfolio S is:

$$\sigma_S^2 = w_S \times \Sigma \times w_S^T$$

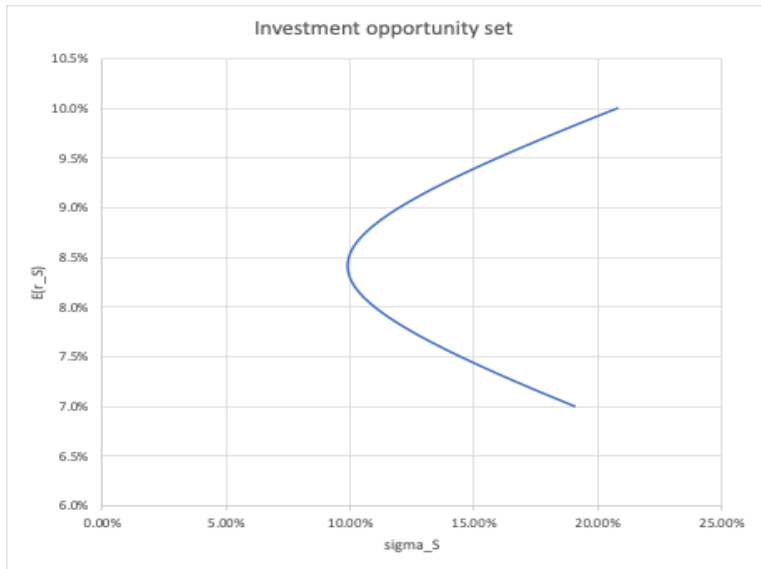
- Return volatility of portfolio S is:

$$\sigma_S = \sqrt{w_S \times \Sigma \times w_S^T}$$

- To find the opportunity set of all portfolios S constructed from assets P and Q , we plot $E(r_S)$ against σ_S

- We illustrate how to build this opportunity set with an example
- Suppose that the two risky assets have the following return statistics:
 - P has $E(r_P) = 8\%$ and $\sigma_P = 11\%$
 - Q has $E(r_Q) = 9\%$ and $\sigma_Q = 12\%$
 - Return covariance between P and Q is $\sigma_{PQ} = \rho \times \sigma_P \times \sigma_Q$, where $\rho = 0.5$ is the *return correlation* between P and Q
- Consider a portfolio S with weights $w_S = (x, 1 - x)$
- The expected return and return volatility of portfolio S become:
 - $E(r_S) = E(r) \times w_S^T = E(r) \times \begin{pmatrix} x \\ 1 - x \end{pmatrix}$
 - $\sigma_S^2 = w_S \times \Sigma \times w_S^T = (x, 1 - x) \times \Sigma \times \begin{pmatrix} x \\ 1 - x \end{pmatrix}$
- Note that both $E(r_S)$ and σ_S^2 depend on x

E(r_P)=	8%		x	1-x	E(r_S)	sigma_S
E(r_Q)=	9%		-1	2	10.0%	20.81%
			-0.9	1.9	9.9%	19.80%
sigma_P=	11%		-0.8	1.8	9.8%	18.81%
sigma_Q=	12%		-0.7	1.7	9.7%	17.84%
rho=	50%		-0.6	1.6	9.6%	16.90%
			-0.5	1.5	9.5%	15.98%
E(r) =	8%	9%	-0.4	1.4	9.4%	15.09%
			-0.3	1.3	9.3%	14.24%
Sigma =	0.0121	0.6600%	-0.2	1.2	9.2%	13.44%
	0.6600%	0.0144	-0.1	1.1	9.1%	12.69%
			0	1	9.0%	12.00%
			0.1	0.9	8.9%	11.39%
			0.2	0.8	8.8%	10.87%
			0.3	0.7	8.7%	10.45%
			0.4	0.6	8.6%	10.14%
			0.5	0.5	8.5%	9.96%
			0.6	0.4	8.4%	9.91%
			0.7	0.3	8.3%	10.00%
			0.8	0.2	8.2%	10.21%
			0.9	0.1	8.1%	10.55%
			1	0	8.0%	11.00%
			1.1	-0.1	7.9%	11.55%
			1.2	-0.2	7.8%	12.18%
			1.3	-0.3	7.7%	12.88%
			1.4	-0.4	7.6%	13.65%
			1.5	-0.5	7.5%	14.47%
			1.6	-0.6	7.4%	15.33%
			1.7	-0.7	7.3%	16.22%
			1.8	-0.8	7.2%	17.15%
			1.9	-0.9	7.1%	18.10%
			2	-1	7.0%	19.08%



- The takeaway from the picture above is that the investment opportunity set generated by two risky assets is a curve (a parabola, to be more exact)
- The two risky assets, P and Q , are also on the investment opportunity set
- The portfolio at the tip of the parabola has the lowest risk, and we call it the *minimum-variance* portfolio
- Note that for any given level of risk, say $\sigma_S = 15\%$, there are two portfolios in the opportunity set that have that level of risk: one on the upper branch, and the other on the lower branch
- A risk-averse investor would always prefer the portfolio on the upper branch to the portfolio with the same risk on the lower branch

- Suppose now that in addition to the two risky assets, P and Q , we can also invest in the risk-free asset, F
- We now want to find the investment opportunity set generated by the two risky assets P and Q and the risk-free asset F
- Suppose that the assets have the following return statistics:
 - P has $E(r_P) = 8\%$ and $\sigma_P = 11\%$
 - Q has $E(r_Q) = 9\%$ and $\sigma_Q = 12\%$
 - F has $E(r_F) = r_F = 2\%$ and $\sigma_F = 0\%$ (because it's risk-free)
 - Return correlation between P and Q is $\rho = 0.5$ and the return covariance $\sigma_{PQ} = \rho \times \sigma_P \times \sigma_Q = 0.0066$
 - Return correlation between F and P and between F and Q are both 0, because F is risk-free
 - Return covariance between F and P and between F and Q are, respectively, $\sigma_{PF} = 0 \times \sigma_P \times \sigma_F = 0$ and $\sigma_{QF} = 0 \times \sigma_Q \times \sigma_F = 0$

- Consider a portfolio S with the weights $w_S = (w_P, w_Q, w_F)$, where $w_P + w_Q + w_F = 1$
- We want to calculate the expected return, $E(r_S)$, and return volatility, σ_S , of portfolio S
- We first rearrange the return statistics in matrix form
- The matrix of expected returns is

$$E(r) = (E(r_P) \quad E(r_Q) \quad E(r_F)) = (8\% \quad 9\% \quad 2\%)$$

- The variance-covariance matrix is:

$$\Sigma = \begin{pmatrix} \sigma_P^2 & \sigma_{PQ} & \sigma_{PF} \\ \sigma_{QP} & \sigma_Q^2 & \sigma_{QF} \\ \sigma_{FP} & \sigma_{FQ} & \sigma_F^2 \end{pmatrix} = \begin{pmatrix} (11\%)^2 & 0.0066 & 0 \\ 0.0066 & (12\%)^2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

- The expected return of portfolio S becomes:

$$E(r_S) = E(r) \times w_S^T = (8\% \quad 9\% \quad 2\%) \times \begin{pmatrix} w_P \\ w_Q \\ w_F \end{pmatrix}$$

- The return variance becomes:

$$\begin{aligned} \sigma_S^2 &= w_S \times \Sigma \times w_S^T \\ &= (w_P \quad w_Q \quad w_F) \times \begin{pmatrix} (11\%)^2 & 0.0066 & 0 \\ 0.0066 & (12\%)^2 & 0 \\ 0 & 0 & 0 \end{pmatrix} \times \begin{pmatrix} w_P \\ w_Q \\ w_F \end{pmatrix} \end{aligned}$$

- Finally, the return volatility, σ_S , becomes:

$$\sigma_S = \sqrt{w_S \times \Sigma \times w_S^T}$$

- As an alternative to the matrix approach of calculating expected returns and variance/volatility, we can also use the more traditional approach:

Expected return of a portfolio of three assets

If portfolio C has a weight of w_P on asset P , w_Q on asset Q , and w_F on asset F , then the expected return of portfolio C can be calculated as the weighted average:

$$E(r_C) = w_P \times E(r_P) + w_Q \times E(r_Q) + w_F \times E(r_F)$$

Variance and volatility of a portfolio of three assets

If portfolio C has a weight of w_P on asset P , w_Q on asset Q , and w_F on asset F , then the variance of portfolio C can be calculated as:

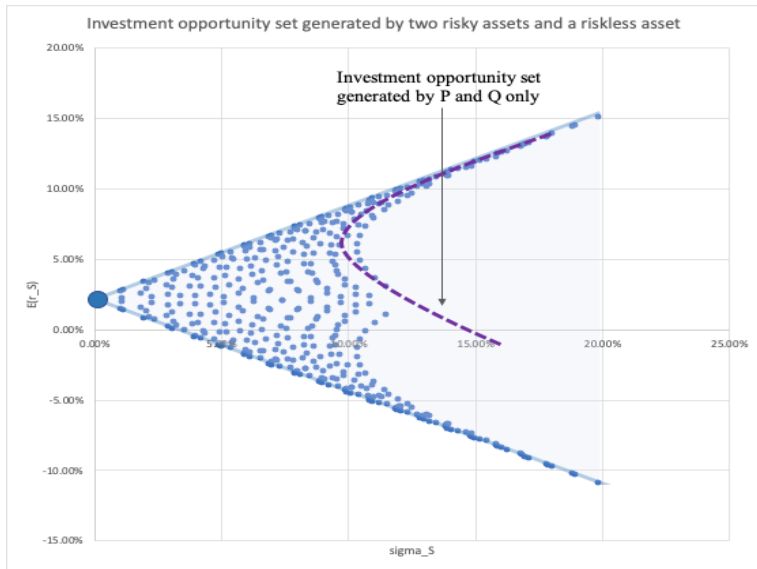
$$\sigma_C^2 = w_P^2 \times \sigma_P^2 + w_Q^2 \times \sigma_Q^2 + w_F^2 \times \sigma_F^2 + 2w_Pw_Q\sigma_{PQ} + 2w_Pw_F\sigma_{PF} + 2w_Qw_F\sigma_{QF} \quad (1)$$

while the volatility of portfolio C is simply the square root of the variance:

$$\sigma_C = \sqrt{\sigma_C^2}$$

- σ_{PQ} is the covariance of asset P and Q ,
- σ_{PF} is the covariance of asset P and F ,
- σ_{QF} is the covariance of asset Q and F

E(r_P)=	8%				w_P	w_Q	w_F	E(r_S)	sigma_S
E(r_Q)=	9%				-1	-1	3	-11.00%	19.92%
r_F	2%				-1	-0.9	2.9	-10.30%	18.88%
					-1	-0.8	2.8	-9.60%	17.85%
sigma_P=	11%				-1	-0.7	2.7	-8.90%	16.85%
sigma_Q=	12%				-1	-0.6	2.6	-8.20%	15.88%
sigma_F=	0%				-1	-0.5	2.5	-7.50%	14.93%
rho=	50%				-1	-0.4	2.4	-6.80%	14.03%
					-1	-0.3	2.3	-6.10%	13.17%
E(r) =	8%	9%	2%		-1	-0.2	2.2	-5.40%	12.38%
					-1	-0.1	2.1	-4.70%	11.65%
Sigma =	0.0121	0.0066	0		-1	0	2	-4.00%	11.00%
	0.6600%	0.0144	0		-1	0.1	1.9	-3.30%	10.45%
	0	0	0		-1	0.2	1.8	-2.60%	10.02%
					-1	0.3	1.7	-1.90%	9.71%
					-1	0.4	1.6	-1.20%	9.55%
					-1	0.5	1.5	-0.50%	9.54%
					-1	0.6	1.4	0.20%	9.68%
					-1	0.7	1.3	0.90%	9.96%
					-1	0.8	1.2	1.60%	10.37%
					-1	0.9	1.1	2.30%	10.90%
					-1	1	1	3.00%	11.53%
					-0.9	-1	2.9	-10.40%	18.99%
					-0.9	-0.9	2.8	-9.70%	17.93%
					-0.9	-0.8	2.7	-9.00%	16.89%
					-0.9	-0.7	2.6	-8.30%	15.87%
					-0.9	-0.6	2.5	-7.60%	14.87%
					-0.9	-0.5	2.4	-6.90%	13.91%
					-0.9	-0.4	2.3	-6.20%	12.98%
					-0.9	-0.3	2.2	-5.50%	12.11%
					-0.9	-0.2	2.1	-4.80%	11.29%
					-0.9	-0.1	2	-4.10%	10.55%
					-0.9	0	1.9	-3.40%	9.90%
					-0.9	0.1	1.8	-2.70%	9.36%
					-0.9	0.2	1.7	-2.00%	8.94%



- The investment opportunity set generated by the two risky assets, P and Q , and the risk-free asset, F , is the blue area spanned by the two lines that stem out of F and that are *tangent* to the investment opportunity set generated by P and Q only
- The two tangent lines are called the *minimum variance frontier*
- For a given level of risk, say $\sigma_S = 15\%$, there are many portfolios in the investment opportunity set with that level of risk
- However, of all the portfolios with $\sigma_S = 15\%$, a risk averse investor would always choose the portfolio on the upper branch of the minimum variance frontier
- This upper branch is known as the *efficient frontier* and is illustrated with the green line
- Thus, risk averse investors are only interested in portfolios on the efficient frontier and completely ignore all other portfolios in the investment set



- Of all the portfolios on the efficient frontier, there is one portfolio that allocates a weight of zero to the risk-free asset F
- This portfolio is denoted with T and invests only in the risky assets, P and Q
- Portfolio T is the tangency point between the efficient frontier and the investment opportunity set generated by P and Q only
- Portfolio T is called the Optimal Risky Portfolio (ORP)

Finding the Optimal Risky Portfolio

The optimal risky portfolio, T , is the portfolio of risky assets (i.e., P and Q only) with the largest Sharpe's ratio

- This can be easily done in Excel
- Geometrically, the Sharpe's ratio is the slope of the line connecting the risk-free asset, F , and portfolio T
- Note that the line connecting F and T is a capital allocation line

- We noticed that a risk-averse investor would only consider investing in the portfolios on the efficient frontier
- The efficient frontier is a line that passes through the risk-free asset, F , and the ORP portfolio, T
- Any portfolio on the efficient frontier can be viewed as a combination of F and T
- In particular, an investor will choose how to allocate its wealth between F and T depending on its risk aversion
- We performed this exact analysis in the case with one risky asset above and we apply it to the risky asset T
- In particular, an investor with risk aversion A will allocate y^* in portfolio T and $1 - y^*$ in the risk-free asset F , where:

$$y^* = \frac{E(r_T) - r_F}{A \times \sigma_T^2}$$

- We now look at the most general case where we want to invest in risky assets $P_1, P_2, \dots$, and P_n , for some $n > 2$
- Just like before, matrices are the easiest way to store the stats information on all these assets
- In particular, the expected returns on these assets are:

$$E(r) = (E(r_{P_1}) \quad E(r_{P_2}) \quad \dots \quad E(r_{P_n}))$$

- Similarly, the variance-covariance matrix stores all the risk information:

$$\Sigma = \begin{pmatrix} \sigma_{P_1}^2 & \sigma_{P_1 P_2} & \dots & \sigma_{P_1 P_n} \\ \sigma_{P_2 P_1} & \sigma_{P_2}^2 & \dots & \sigma_{P_2 P_n} \\ \dots & \dots & \dots & \dots \\ \sigma_{P_n P_1} & \sigma_{P_n P_2} & \dots & \sigma_{P_n}^2 \end{pmatrix}$$

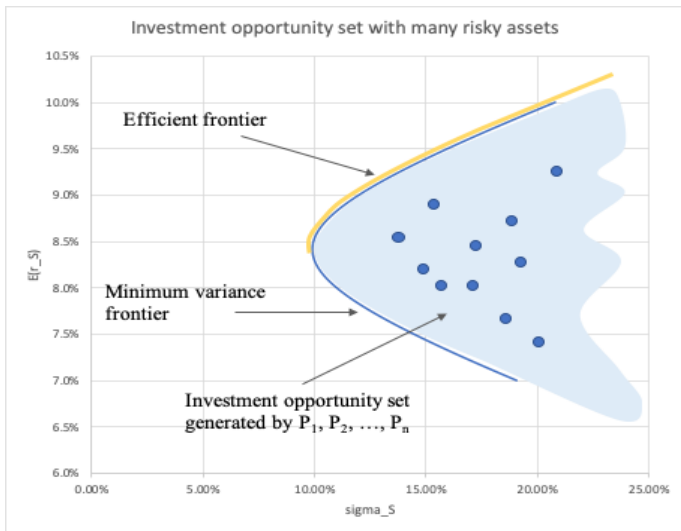
- Consider portfolio S with weights $w_S = (w_{P_1}, w_{P_2}, \dots, w_{P_n})$
- The expected return of portfolio S is:

$$E(r_S) = E(r) \times w_S^T$$

- The return volatility portfolio S is:

$$\sigma_S = \sqrt{w_S \times \Sigma \times w_S^T}$$

- The investment opportunity set of all portfolios S that invest in the risky assets $P_1, P_2, \dots, P_n$ is an area bounded by a parabola
- The parabola on the outside of this area is called the *minimum variance frontier* generated by the assets $P_1, P_2, \dots, P_n$
- For a given level of risk, investors would always want to be on the upper half of the minimum variance frontier, which is called the *efficient frontier* generated by the assets $P_1, P_2, \dots, P_n$



- Suppose now that investors could also invest in a risk-free asset, F , in addition to the risky assets $P_1, P_2, \dots$, and P_n , for some $n > 2$
- As usual, we denote the expected return of F with r_F and adjust the expected returns of all assets:

$$E(r) = (E(r_{P_1}) \quad E(r_{P_2}) \quad \dots \quad E(r_{P_n}) \quad r_F)$$

- Similarly, because F has no risk, we adjust the variance-covariance matrix by adding 0's for all entries associated with F :

$$\Sigma = \begin{pmatrix} \sigma_{P_1}^2 & \sigma_{P_1 P_2} & \dots & \sigma_{P_1 P_n} & 0 \\ \sigma_{P_2 P_1} & \sigma_{P_2}^2 & \dots & \sigma_{P_2 P_n} & 0 \\ \dots & \dots & \dots & \dots & 0 \\ \sigma_{P_n P_1} & \sigma_{P_n P_2} & \dots & \sigma_{P_n}^2 & 0 \\ 0 & 0 & \dots & 0 & 0 \end{pmatrix}$$

- Consider a portfolio S that invests in the n risky assets $P_1, P_2, \dots, P_n$
- Suppose that the weights of portfolio S are

$$w_S = (w_{P_1} \quad w_{P_2} \quad \dots \quad w_{P_n})$$

- Just like before, the expected return of portfolio S is:

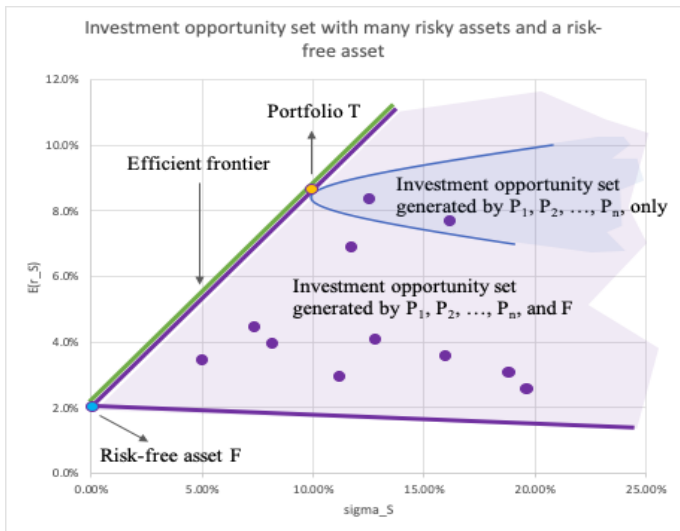
$$E(r_S) = E(r) \times w_S^T$$

- And the return volatility portfolio S is:

$$\sigma_S = \sqrt{w_S \times \Sigma \times w_S^T}$$

- The investment opportunity set of all portfolios S is an area bounded by a parabola that includes all the risky assets $P_1, \dots, P_n$
- The parabola that borders this area has the property that for a given expected return the portfolio on the border that has this expected return also has the lowest variance
- As a result, this bordering parabola is called the *minimum-variance frontier*
- Suppose that we now want to invest in all risky assets $P_1, P_2, \dots, P_n$ as well as the risk-free asset F
- What is the investment opportunity set of all portfolios S consisting of all these assets?

- Similar to the case when we only had two risky asset, the investment opportunity set is an area bounded by two lines that stem from the risk-free asset F and that touch tangentially the minimum-variance frontier generated by the risky assets $P_1, P_2, \dots, P_n$
- Investors like only the portfolios on the upper line
- We call this line the *efficient frontier* generated by the risk-free asset F and risky assets $P_1, P_2, \dots, P_n$
- Note that the efficient frontier is tangent to the minimum-variance frontier generated by the risky assets $P_1, P_2, \dots, P_n$ only
- The portfolio at the tangency point is denoted with T and consists of $P_1, P_2, \dots, P_n$ only (no risk-free asset)
- Portfolio T is called the optimal risky portfolio (ORP)



- Just like in the case with two risky assets, the slope of the efficient frontier is the Sharpe ratio
- In particular, the ORP portfolio, T , is the portfolio of $P_1, \dots, P_n$ with the highest Sharpe's ratio
- Using our notation, we can calculate the Sharpe's ratio of a portfolio S with weights w_S as follows:

$$SR_S = \frac{E(r_S) - r_F}{\sigma_S} = \frac{E(r) \times w_S^T - r_F}{\sqrt{w_S \times \Sigma \times w_S^T}}$$

- The portfolio weights w_S^* that maximizes the right-hand side of the above expression are exactly the weights of our ORP portfolio, T

- All investors choose to hold portfolios on the efficient frontier, i.e., the line connecting the risk-free asset, F , and the ORP portfolio T
- How much each investor choose to hold of either asset depends on their risk aversion, A
- In particular, an investor with risk aversion A will allocate y^* to the tangent portfolio T and $1 - y^*$ to portfolio F , where:

$$y^* = \frac{E(r_T) - r_F}{A \times \sigma_T^2}$$

- The portfolio with weights y^* in ORP and $1 - y^*$ in F is called the investor's **optimal complete portfolio**

- The portfolio analysis above uncovers a very interesting result:

One-fund separation result

All risk-averse investors choose to hold the risk-free asset, F , and the *same* portfolio of risky assets, namely the ORP portfolio, T . The mix of F and T that each investor holds depends on its risk aversion.

- The analysis above pins down quite precisely the optimal portfolio allocation of any risk averse investor
- But implementing this portfolio in reality may be challenging for a number of reasons:
 - Risk premium may not be estimated with precision for some assets (e.g., not enough data points in the time series)
 - The estimation of the variance-covariance matrix relies on high frequency data (e.g., daily returns) and can be technical and time consuming
 - The weights of the ORP portfolio T change over time, which means that investors have to *actively manage* this portfolio (i.e., costly)
 - An investor's risk-aversion coefficient is usually hard to pin down with precision

- Instead of going to all this trouble, investors could view the main implication of the theory above as saying that the best an investor can do is to hold the risk-free asset and a well-diversified portfolio of risky assets
- In other words, investors would replace the ORP portfolio, T , with a market index, such as S&P 500
- In this case, the efficient frontier is replaced by the line connecting the risk-free asset and the market index
- This line is known as the *capital market line* (CML)
- Investment strategies that allocate capital to the risk-free asset and only readily-available mutual funds or ETFs that track a market index are known as *passive strategies*
- The theory above suggests that passive strategies could be an optimal for some investors